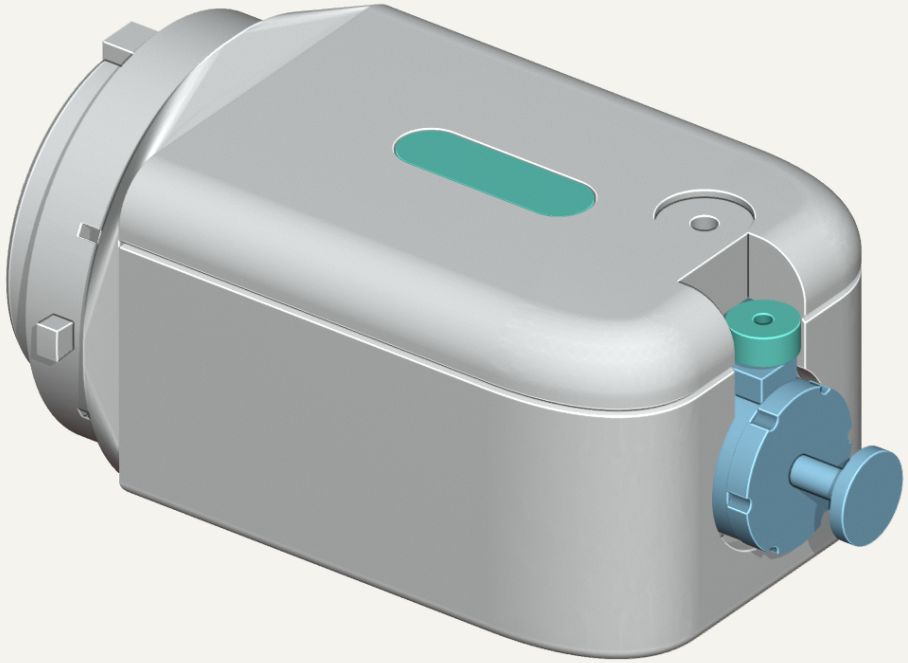
Assemble the R08 head



01 This guide matches the R08 head and the 14-part A1 mini print projects, including the reprint of 42, 47, 49 and 53. Use the actual part shapes and numbers to identify your prints.

02 Start with the completed prints. Remove supports, collect the purchased hardware, then assemble in the order shown. Illustrations use the released CAD; they are not photographs of your build.

CHECK **Colours identify parts. Your printed colours may differ.**

R08 / HEAD ONLY / PAUSE AT EACH STEP

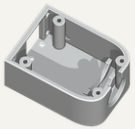
Lay out the printed parts

41



Joint adaptor

42



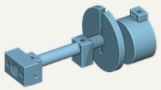
Open chassis

43



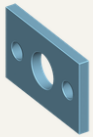
Removable cover

44



Moving carriage

46



Magnet retainer

47



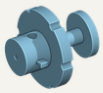
Hall mount

48



Filament cartridge

49



Bench cartridge

50



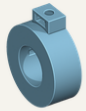
Service knob

51



Cartridge-lock knob

53



Receiver fit coupon

OUTSIDE THE HEAD

55



Temporary bridge

OUTSIDE THE HEAD

56



Optional inlay

57



Printed flexure

Each part is scaled separately for identification. Colours are illustrative.

Clear support residue from pockets and cartridge tracks. Inspect both thin beams before assembly.

Fit-test cartridge 48 in coupon 53. Use only one cassette; 57 / 58 / 59 are not force grades.

Count the hardware first

FASTENER	QTY	LOCATION
M2 x 8 socket-cap	**4**	2 moving rail + 2 fixed rail
M2 x 10 socket-cap	**2**	Magnet retainer
M2 x 12 socket-cap	**1**	Cartridge lock
M2 hex nut / AF4 mm	**7**	4 rail + 2 magnet + 1 lock
M3 x 6 socket-cap	**2**	Hall mount
M3 x 8 button-head	**4**	Joint adaptor
M3 x 30 socket-cap	**1**	Removable service screw
M3 x 35 socket-cap	**4**	Cover; borrow 2 for bridge
M3 hex nut / AF5.5 mm	**14**	11 in head + 3 in bridge

M3 x 8 button heads: diameter <= 6.4 mm; height <= 2 mm.

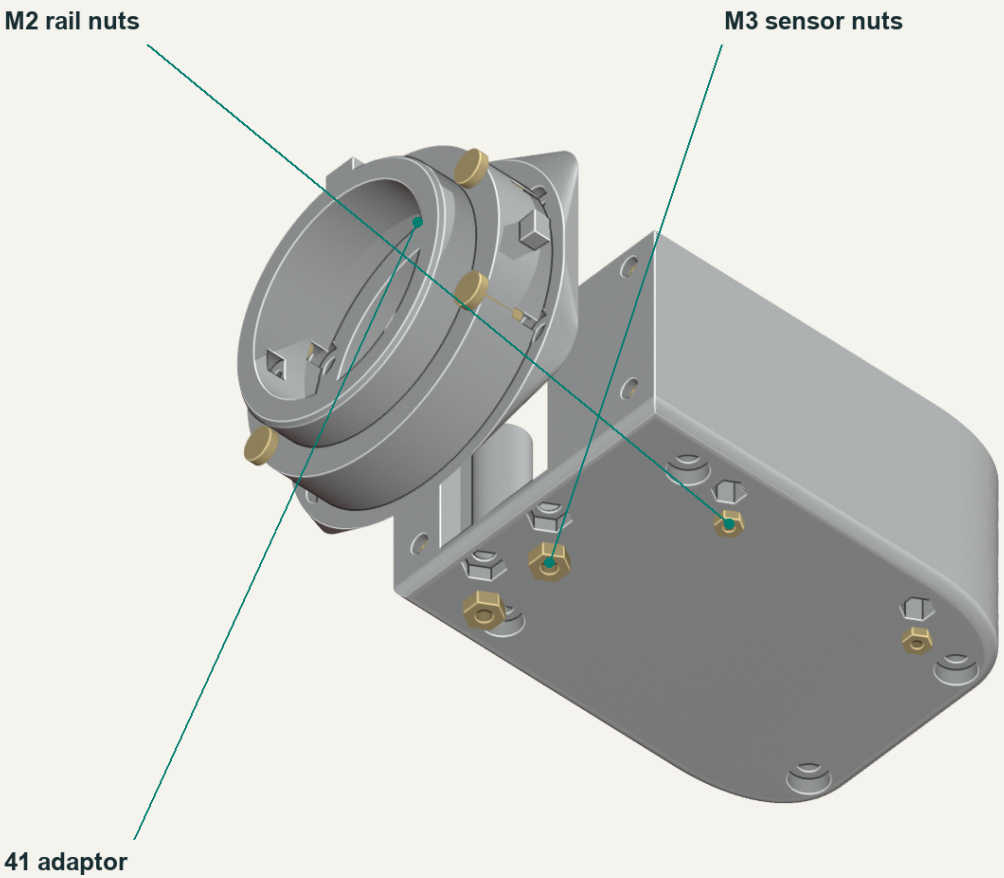
Collect the screws in the table. The 14 M3 nuts include three that remain in the temporary bridge; 11 are used in the head. Borrow two of the four M3 x 35 cover screws to hold the bridge.

Use small hand tools and tighten gently. Prefer brass hardware around the magnet. Never use screw torque to pull a misaligned part into place.

You also need a 6 x 3 mm axially magnetized disc, a DRV5055A3QLPG sensor, insulated leads and a 100 nF capacitor. The first electrical test uses external 3.3 V bench electronics.

Tools: M2/M3 hex drivers, tweezers, calipers, compatible adhesive, soldering tools and heat-shrink.

Prepare the chassis



01 Fit four M3 nuts in the rear adaptor pockets, two M3 nuts in the underside sensor pockets and two M2 nuts in the deep underside cassette pockets.

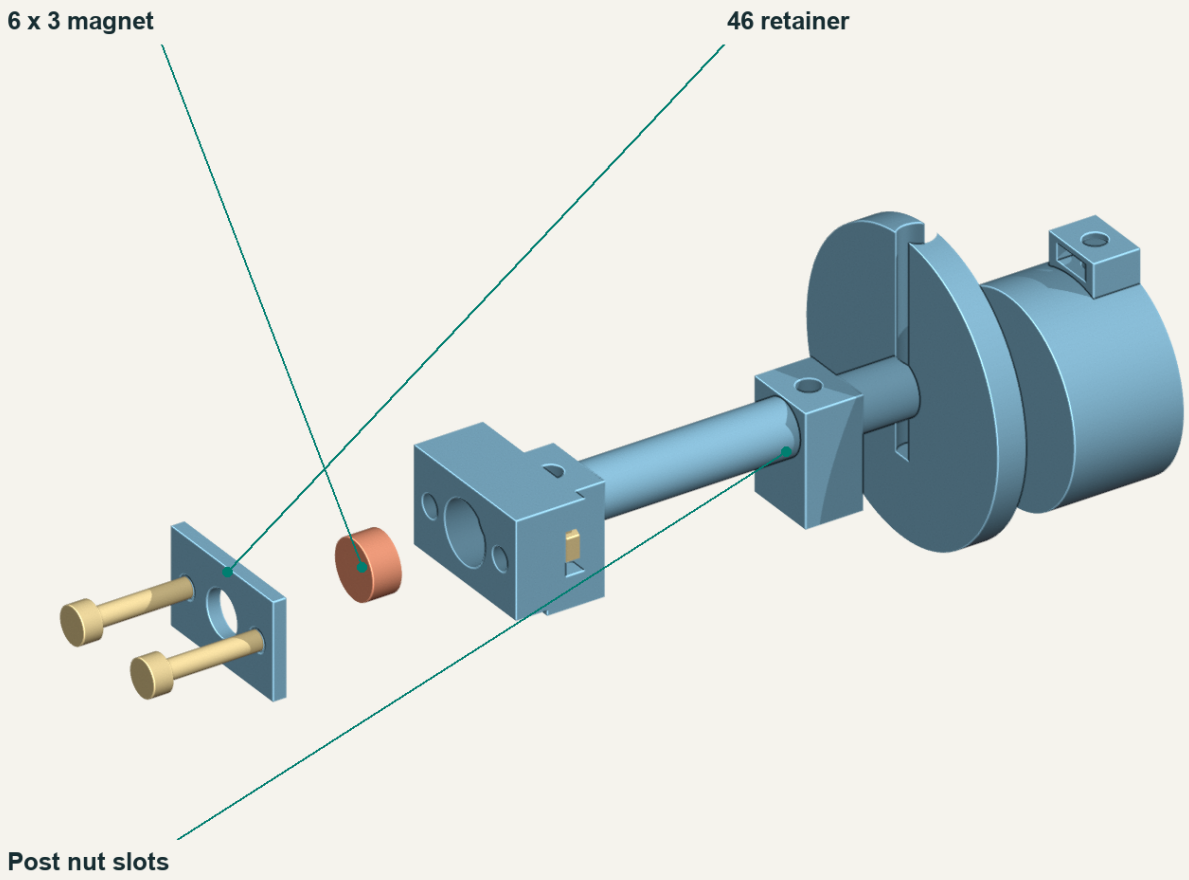
02 A tiny compatible adhesive tack can hold the deep cassette nuts. Keep the threads clear and let it cure before assembly.

03 Join adaptor 41 using four M3 x 8 button-head screws. Taller socket-cap heads foul the joint counterbores. The underside view shows the pocket locations.

CHECK **Nuts sit fully in their pockets; adaptor faces meet without force.**

4 x M3 x 8 BUTTON-HEAD / 6 x M3 NUTS + 2 x M2 NUTS

Build the carriage on the bench



Hidden nut pockets: see page 15.

01 Before putting the carriage in the housing, slide two M2 nuts into its downward-post slots and two more M2 nuts into the magnet-retainer slots.

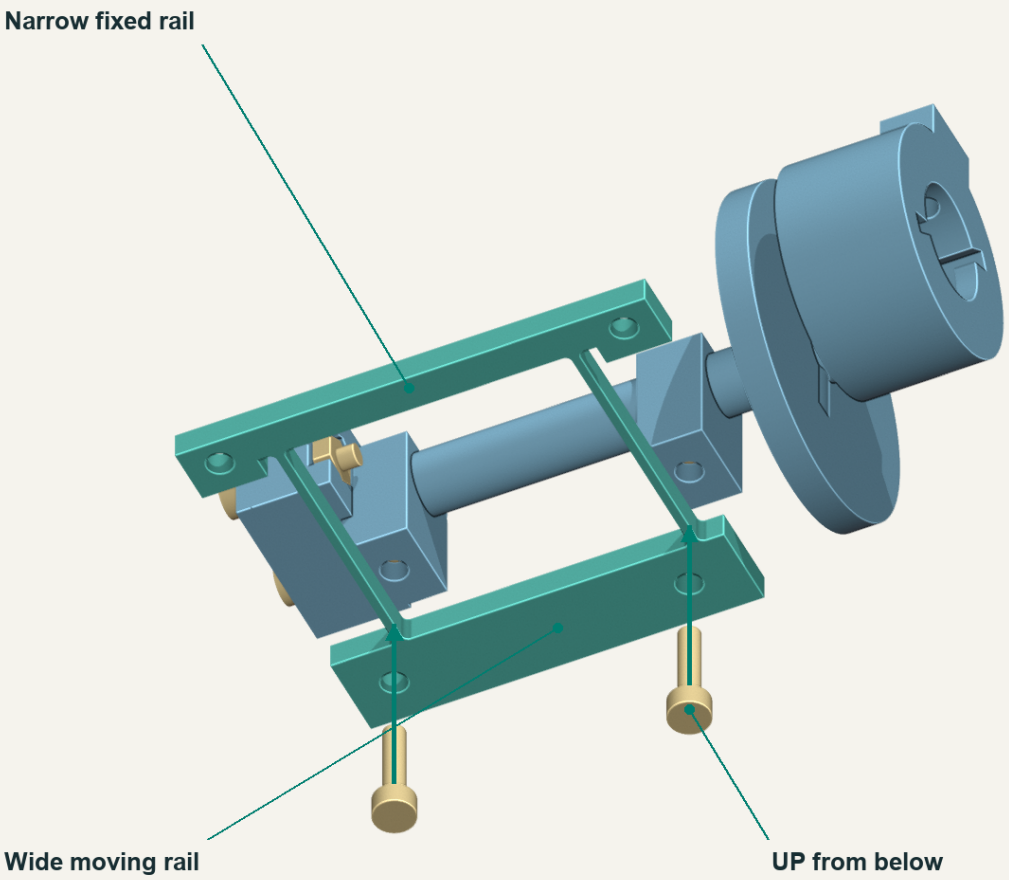
02 Use a thin compatible adhesive layer to secure the axially magnetized 6 x 3 mm disc in its pocket. Choose a pole orientation and keep it fixed. Let the adhesive cure.

03 Fit cap 46 over the magnet with two M2 x 10 screws. Keep adhesive out of the screw threads and away from the cassette mounting faces.

CHECK **Magnet cannot rattle; post faces and nut slots stay clean.**

4 x M2 NUTS / 2 x M2 x 10 / 6 x 3 mm MAGNET

Attach the moving rail from below



01 Orient the cassette with its wide moving rail against the bottoms of the two carriage posts. The narrow fixed rail is on the opposite side of the two thin beams.

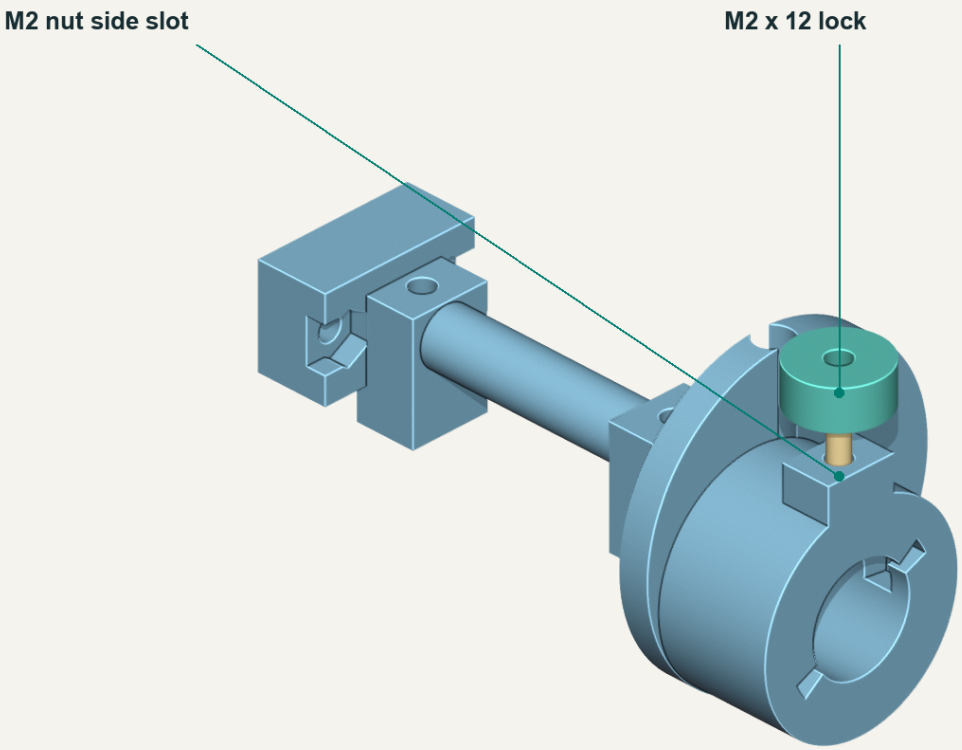
02 Insert two M2 x 8 screws upward from beneath the cassette into the post nuts. Their heads remain underneath. Tighten evenly without crushing the plastic.

03 Leave the opposite rail free. Keep both flexible beams and all mounting faces free of adhesive. Do not bend the cassette to make holes line up.

CHECK **Do this outside the chassis. The floor blocks later driver access.**

2 x M2 x 8 SOCKET-CAP / EXISTING POST NUTS

Fit the small cartridge lock



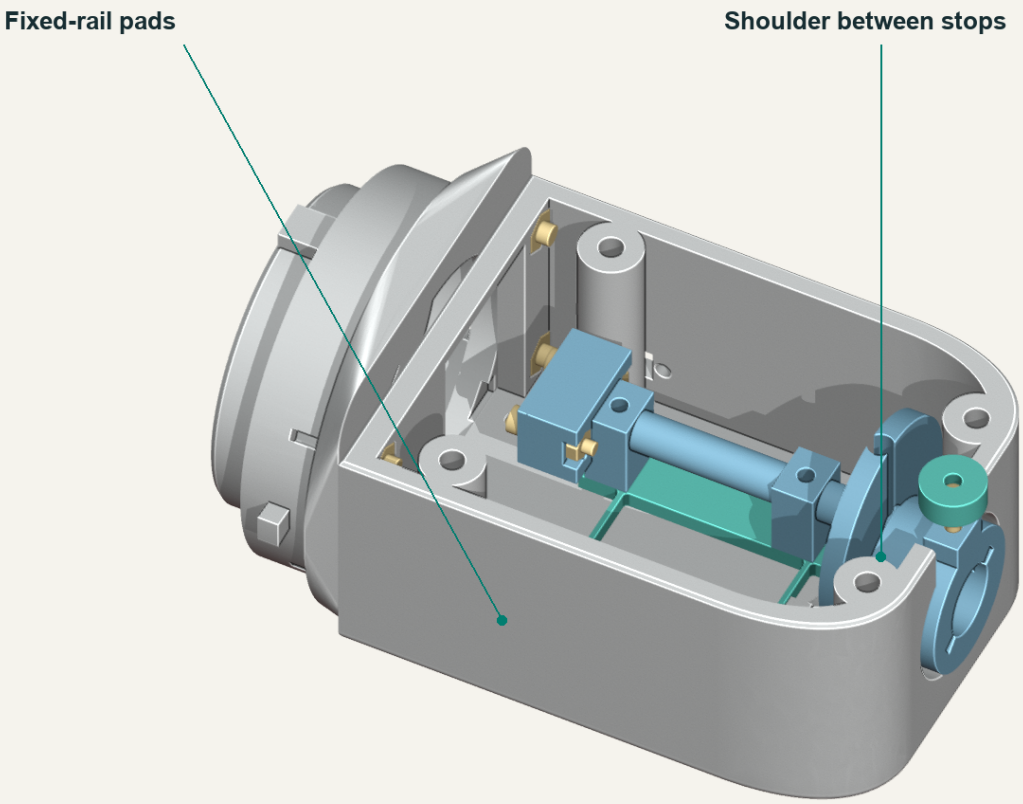
01 Seat one M2 nut in the side-loading slot at the cartridge receiver. Fit the M2 x 12 screw through the top boss.

02 Optional knob 51 bonds over the screw head. Leave the lock backed out while fitting a cartridge; the screw tip must clear the cartridge path.

CHECK **This is the small cartridge lock, not the long service screw.**

1 x M2 NUT / 1 x M2 x 12 / OPTIONAL KNOB 51

Lower the complete subassembly



01 Lower the carriage and attached cassette together into the open chassis. Hold the solid carriage or thick rail; never lift or pull by the beams.

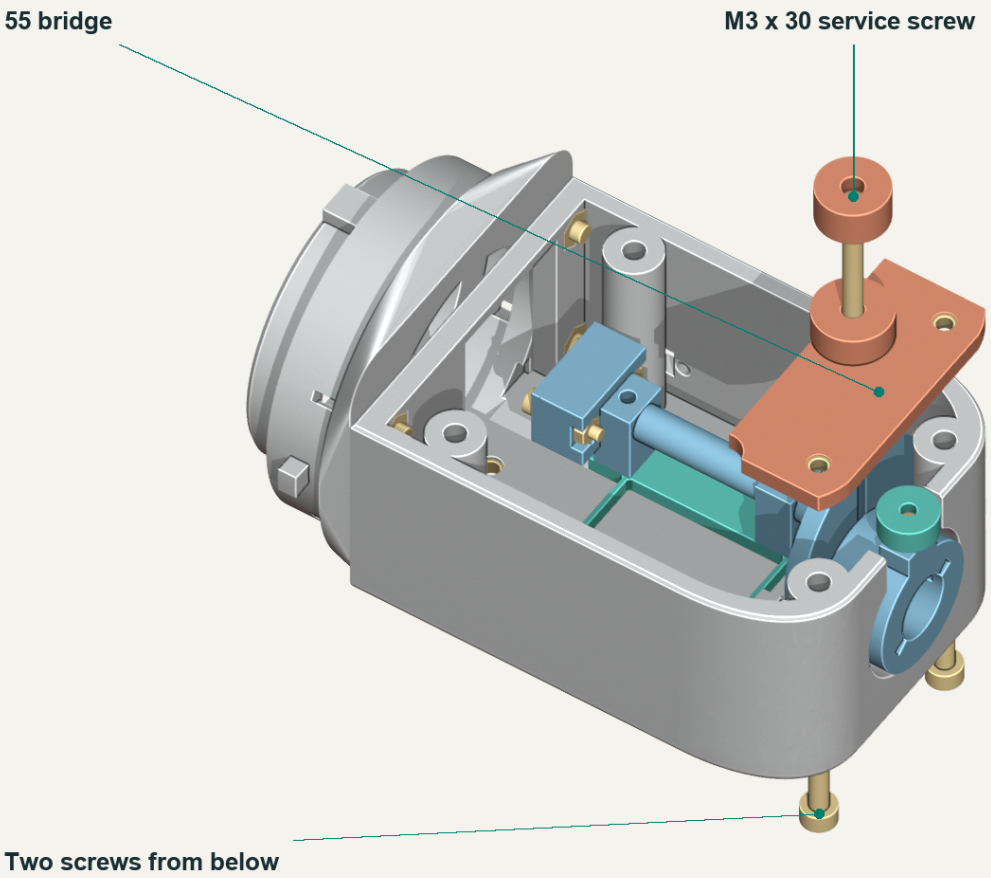
02 The narrow fixed rail rests on the two pads beside the long side wall. The large carriage shoulder must lie between the lower travel stops.

03 Leave the fixed-rail screws out until the temporary bridge locates the carriage. If holes do not align, lift out and check orientation or support residue.

CHECK **Both fixed-rail holes align naturally. Do not force the flexure.**

NO NEW FASTENERS

Locate the carriage with bridge 55



Hidden nut pockets: see page 15.

01 Fit three M3 nuts into the bridge: two for its mounting holes and one for the service screw. Place bridge 55 across the front of the chassis.

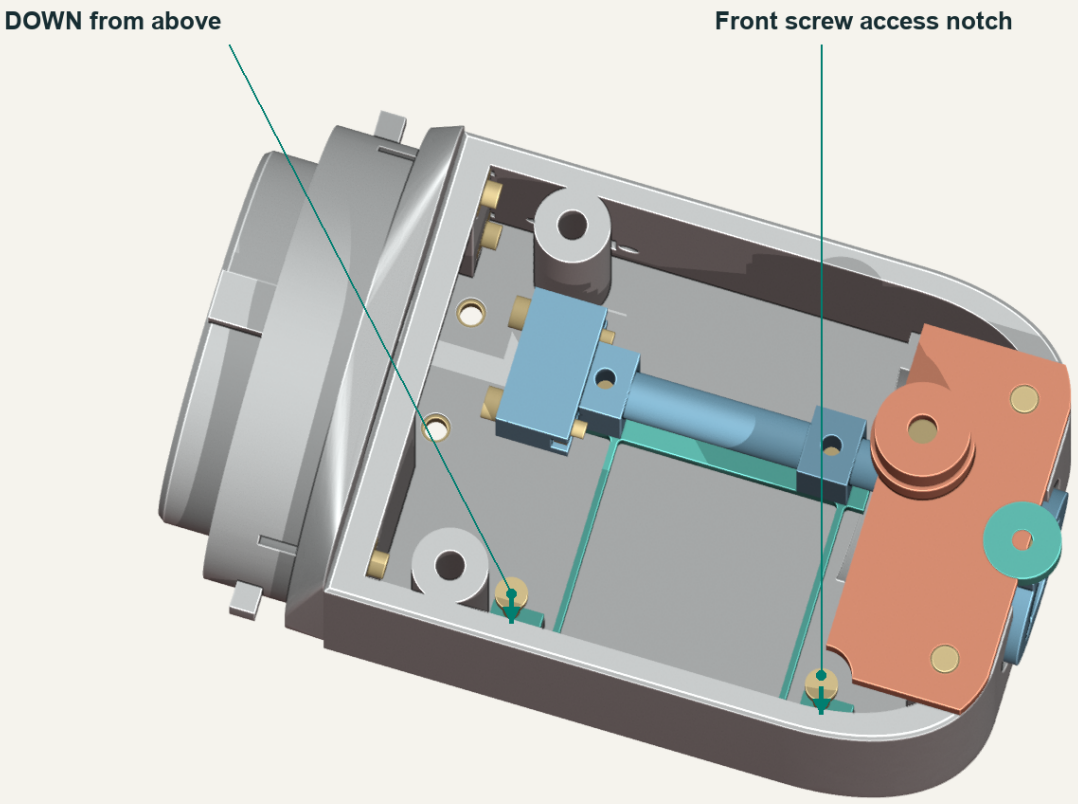
02 Borrow the two front M3 x 35 cover screws. Insert them from below through the chassis into the bridge nuts.

03 Fit the M3 x 30 service screw into the carriage shoulder bore. Optional knob 50 bonds over its head. It holds the carriage near neutral with clearance. Do not use it to force alignment or clamp the carriage down.

CHECK **The service screw enters the shoulder bore with clearance.**

3 x M3 NUTS / 2 x M3 x 35 / 1 x M3 x 30

Secure the fixed rail from above



01 With the bridge locating the carriage, insert two M2 x 8 screws downward through the narrow rail into the chassis nuts. The bridge notch leaves the front screw accessible.

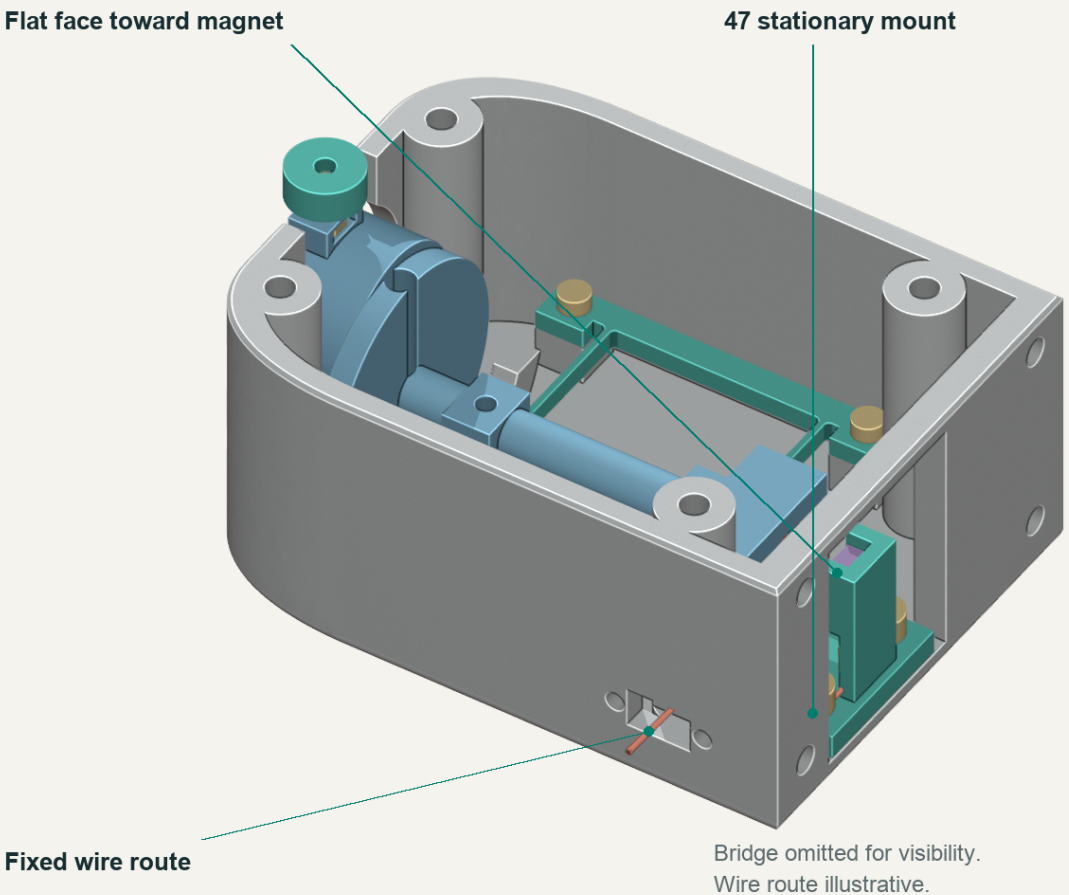
02 Tighten evenly while the cassette remains flat and unstrained. There is no specified tightening torque for this printed prototype; stop before distorting the plastic.

03 Remove the entire service screw and check gentle movement and free return. If the rail bows or the carriage rubs, stop and inspect the mounting faces and alignment.

CHECK **No bowing, rubbing or permanent offset after release.**

2 x M2 x 8 SOCKET-CAP / CHASSIS NUTS ALREADY FITTED

Fit the stationary Hall sensor



Bridge omitted for visibility.
Wire route illustrative.

Sensor close-up and bench wiring: see page 16.

01 With power disconnected, solder insulated leads and the 100 nF VCC-to-GND capacitor before seating the sensor. Verify the TI LPG package pin drawing before soldering.

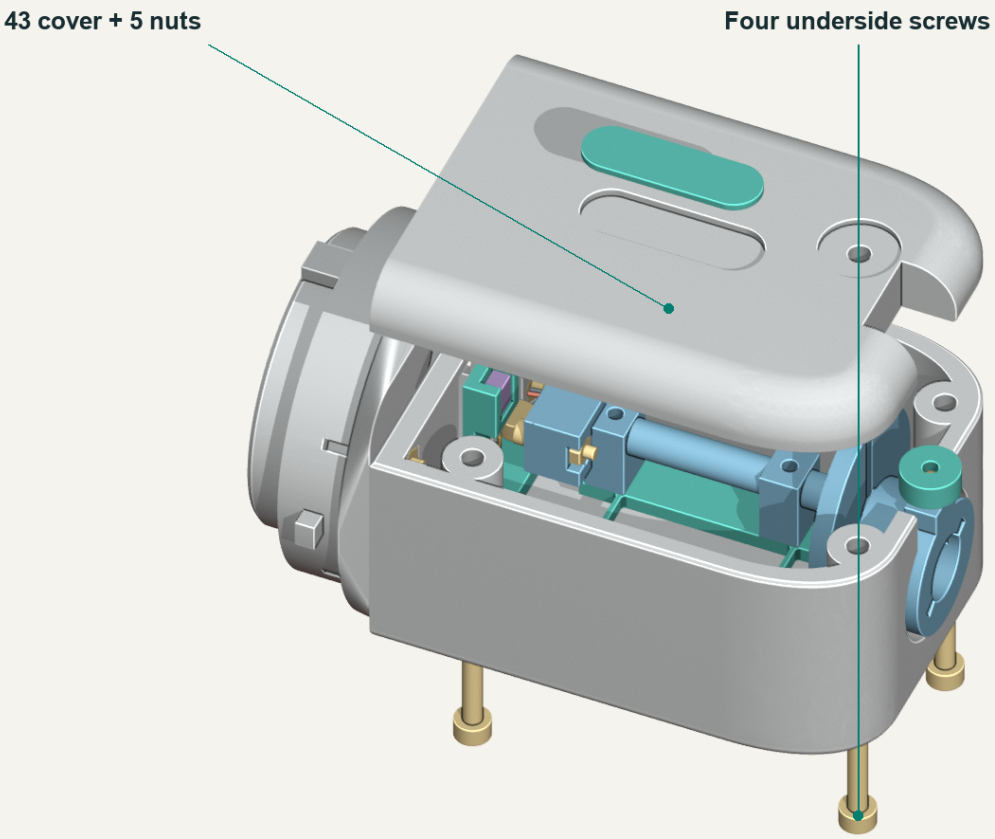
02 Fit the sensor with its flat face toward the magnet. Retain it with a small adhesive fillet on its back or sides, away from the leads and moving parts.

03 Attach mount 47 with two M3 x 6 screws at the middle of its slots. Route and strain-relieve wires on fixed structure through the side port. Nominal package-face gap is 8.2 mm; it is not a force calibration.

CHECK **No wire or adhesive touches the carriage or flexure beams.**

DRV5055A3QLPG / 100 nF CAPACITOR / 2 x M3 x 6

Exchange the bridge for the cover



01 Remove the service screw and the two bridge mounting screws. Lift the bridge out and set it aside with its three nuts. Reuse the two long screws for the cover.

02 Fit four M3 cover nuts plus one M3 service-screw nut in the underside pockets of cover 43. Check the wires are clear of the closing faces.

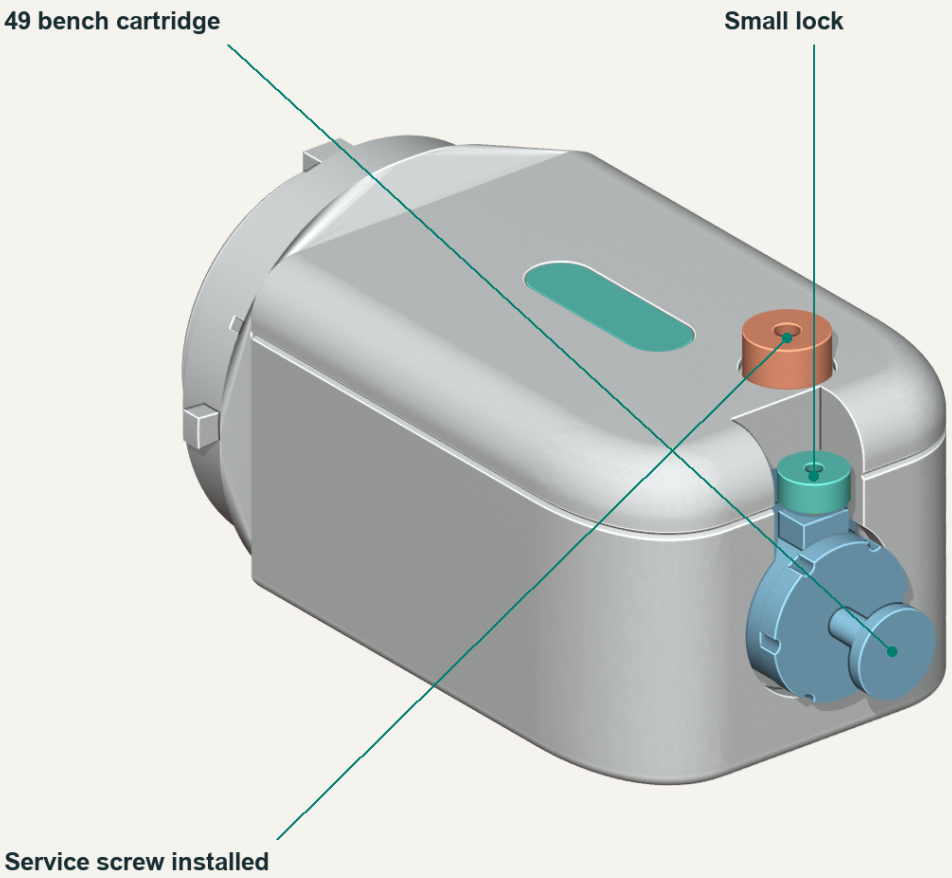
03 Lower the cover and secure it from below with all four M3 x 35 screws. Tighten gradually and evenly. Bond optional inlay 56 in its shallow recess.

Hidden nut pockets: see page 15.

CHECK **Set bridge 55 and its three nuts aside. Check wire clearance.**

5 x M3 NUTS / 4 x M3 x 35 / OPTIONAL INLAY 56

Fit the rigid bench cartridge first



01 With the tip unloaded, install the M3 x 30 service screw to support the carriage. Back the M2 cartridge lock out at least 3 mm so its tip clears the path.

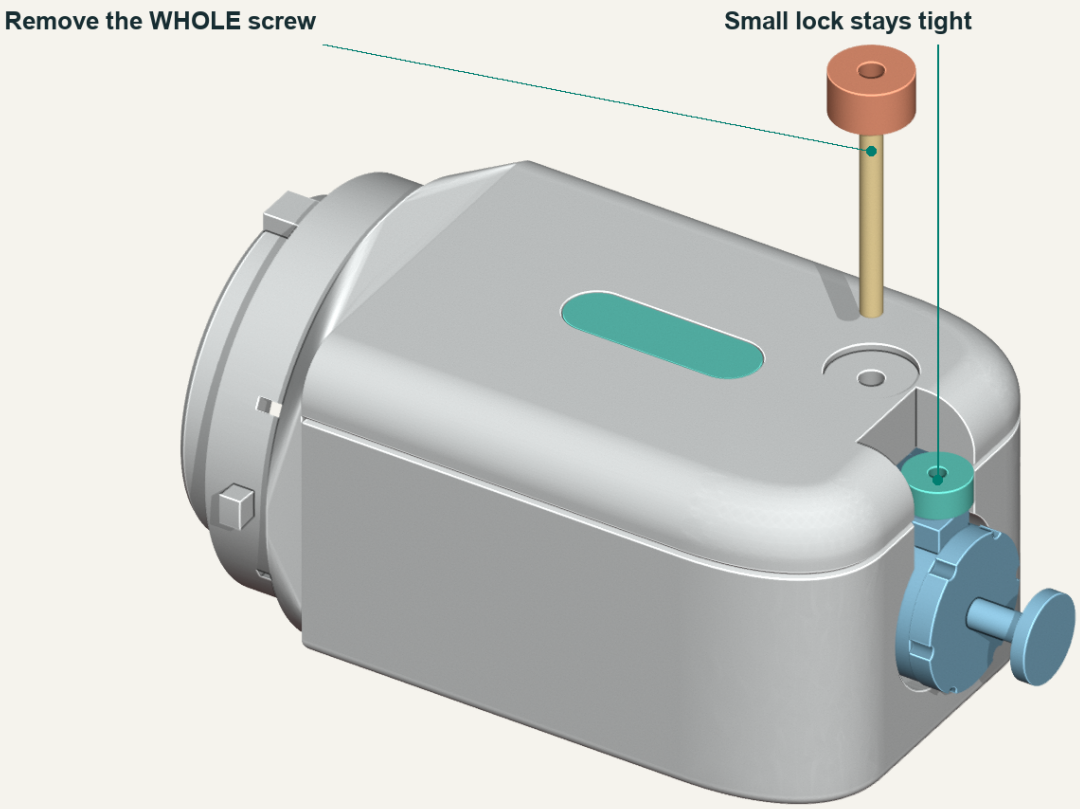
02 Align the cartridge lugs with the receiver entries, insert cartridge 49 and rotate 30 degrees into the locked position. Hold the hub and seat its flange against the receiver face.

03 Tighten the small lock gently. Use the same sequence for cartridge 48 later. Characterize the actual nylon filament separately; its CAD dimensions are not a calibrated force grade.

CHECK **Hold the cartridge hub. Never turn it by pulling the tip.**

49 BENCH CARTRIDGE / EXISTING LOCK + SERVICE SCREW

Remove the service screw. Test return.



01 Withdraw the whole service screw, not just the red knob. The small M2 cartridge lock remains tightened. Keep the bridge outside the head.

02 With cartridge 49 fitted, check gentle axial movement and free return. Do not drive the mechanism against its stops. If it sticks, reopen the head and check supports, wire contact, rail flatness and screw alignment.

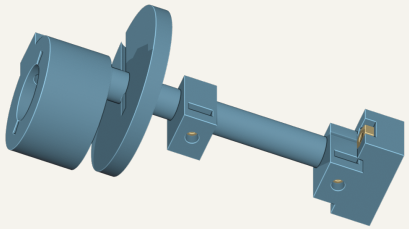
03 Then use external 3.3 V electronics and a checked reference balance. Calibrate the actual cassette and final hardware. Replacing or retightening a cassette requires calibration again. Store unloaded with the service screw removed.

CHECK **Assembly is complete only when movement and return are free.**

SERVICE SCREW OUT / BRIDGE OUT / BENCH CARTRIDGE IN

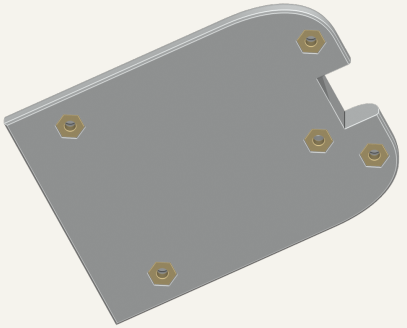
Hidden nut pockets

CARRIAGE 44



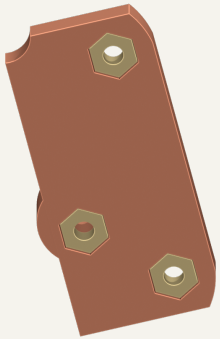
2 M2 post nuts + 2 M2 magnet-retainer nuts.

COVER 43



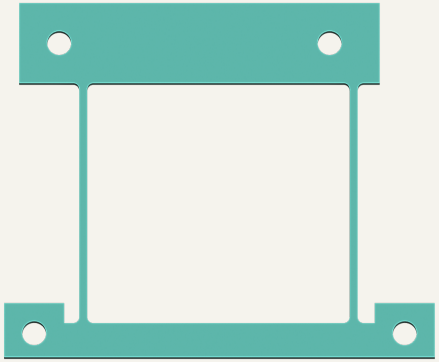
4 M3 cover nuts + 1 M3 service-screw nut.

BRIDGE 55



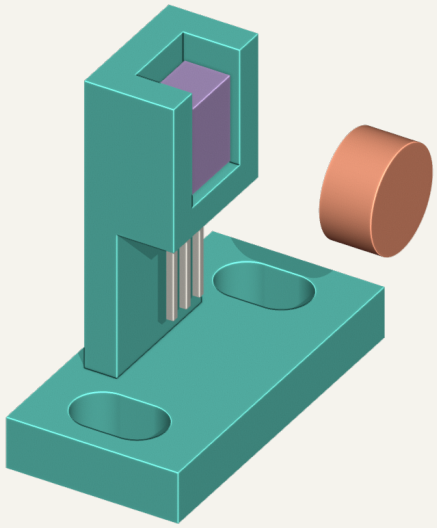
2 M3 mounting nuts + 1 M3 service-screw nut.

CASSETTE 57



Wide rail: carriage. Narrow rail: chassis. Protect both thin beams.

Sensor orientation and bench wiring



The flat TO-92 package face points toward the magnet. The figure uses a package envelope; confirm the real sensor and TI pin drawing before soldering.

Nominal package-face gap: 8.2 mm at the centre of the slots. Adjustment is about 7.2-9.2 mm. Changing the gap or magnet orientation requires calibration again.

Pin numbers above are for TI LPG / TO-92. TI DRV5055 datasheet, pin drawing (page 3).

Hall pin 1 - VCC

Regulated 3.3 V

Hall pin 2 - GND

Common ground

Hall pin 3 - OUT

ADS1115 A0

100 nF capacitor

Between VCC and GND, near sensor

ADS1115 supply / GND

Same 3.3 V rail / common ground

ADS1115 SDA / SCL

Controller I2C; pull-ups to 3.3 V

Connect with power off. Insulate the leads, strain-relieve the cable and keep all wires clear of the moving stage. Initial electronics remain external.

Optional filter: Hall OUT through 4.7 kohm to A0; 2.2 uF from A0 to GND. Calibrate with the final filter fitted.

Before the first force measurement

- ☐ Bridge 55 is out of the head.
- ☐ The entire M3 x 30 service screw is removed.
- ☐ Cartridge 49 is seated and its small lock is tightened.
- ☐ Cassette is flat, with no cracks or permanent bowing.
- ☐ No wire, adhesive or support residue touches moving parts.
- ☐ Carriage moves gently and returns freely.
- ☐ Cover is secure and no wires are trapped.

Store unloaded with the service screw removed. Record the cassette number, filament, print settings and test temperature.

It sticks or fails to return

Check support residue, pin removal, wire contact, rail flatness and alignment. Loosen and reseat only while unloaded.

A nut spins or will not seat

Clear the pocket and check actual nut size. Do not force the screw or enlarge the whole part by scaling.

A cartridge will not turn

Unload, fit the service screw and back the small lock out at least 3 mm. Check the lugs and tracks in coupon 53.

The spring cracks or stays bent

Replace the cassette. Repeat return checks and recalibrate. Do not sand its beams to tune stiffness.

Calibration handoff and source notes

1. Establish a baseline

Use bench cartridge 49 and a checked reference balance. Zero in the test orientation. Confirm a smooth, unsaturated sensor response.

2. Measure before assigning a force

Record loading and unloading at 0, 1, 2, 5, 10, 20, 50 and 75 gf only while below the stops and within sensor range. Skip unreachable points; do not force them. Repeat at least five cycles.

3. Check drift and return

Follow the R08 calibration guide for hold, recovery and repeated-cycle tests. Recalibrate after changing or retightening the cassette, changing magnetic hardware, or adjusting the sensor.

This is an experimental bench prototype. Assembly does not establish force accuracy, spring fatigue life or patient-contact suitability. Use inanimate fixtures and reference loads for initial tests.

R08 BUILD_GUIDE.md

Assembly order, screw directions and service sequence.

R08 SHOPPING_LIST.md

Fastener counts, real magnet and sensor specification.

R08 assembly_manifest.json + reference STLs

The actual part geometry and final assembly coordinates.

R08 WIRING_AND_CALIBRATION.md

Bench connections, reference loads, return and drift tests.

R08 printing notes, 21 October 2026

The 14-part plate and 42/47/49/53 reprint identification.

Illustrative colours and exploded offsets aid identification. Fasteners are simplified CAD envelopes. Animation is not a tool-access, collision or force simulation. Wire routing is schematic.

Texas Instruments DRV5055 datasheet, Rev. C
Pin functions checked 22 October 2026. The companion source folder records file hashes for the local design references.